

The Formula E to Consumer (FE2C) Optimization Engine

This project focuses on **Energy Efficiency Ratios (EER)**, measuring how teams convert limited battery "fuel" into performance, and then modeling how that efficiency maps to consumer EV challenges (like range anxiety and charging infrastructure in California).

Week 1: Data Engineering & The "Digital Twin" Schema

Instead of a flat results table, build a relational database that treats a race car like a "mobile laboratory."

- **The Task:** Scrape and integrate three distinct data sources:
 1. **Formula E Performance Data:** Lap times, energy consumption per lap (where available), and regen (braking) efficiency.
 2. **Environmental Data:** Pull API data for tracking elevation changes and weather.
 3. **Market Data:** California New Vehicle Registration data (by powertrain/brand).
- **The SQL Challenge:** Build a **Star Schema** where the "Fact" is a specific race stint, and the "Dimensions" include the battery hardware version, the track surface type, and the manufacturer.

Week 2: The "Efficiency Frontier" Analysis (Business Logic)

Toyota is famous for *Kaizen* (continuous improvement) and Lean manufacturing. Your project should analyze which teams are the most "efficient" rather than just the "fastest."

- **Advanced SQL & Python:** Use Window Functions to calculate **Delta-E** (energy used vs. positions gained).
- **The Metric:** Create a proprietary "Toyota Efficiency Score" for every manufacturer.
 - *Calculation:* $(\text{Total Race Distance} / \text{Energy Used}) * (\text{Average Lap Velocity})$.
- **Why it takes time:** You'll need to normalize this across different seasons (Gen 2 vs. Gen 3 cars), which requires complex data cleaning and mapping of technical specifications.

Week 3: The "California Simulation" (The TRI Hook)

This is the part that makes it a "large" project. You will use your Formula E findings to solve a "real-world" Toyota problem.

- **The Scenario:** If Toyota were to launch a high-performance EV in the California market, what battery management strategy (learned from Formula E) would maximize range in high-traffic/high-heat ZIP codes (like the Inland Empire or Central Valley)?
- **The Model:** Build a Python simulation that takes Formula E "Regen" (energy recovery) data and applies it to California traffic patterns (Stop-and-go on the I-405).

- **The Business Output:** A data-driven white paper (or Dashboard) recommending whether Toyota should prioritize **Battery Density** (larger batteries) or **Recovery Efficiency** (faster charging/regen) based on current adoption trends in California.

How to make it "AI-Proof" and Multi-Week

To ensure this takes significant time and shows deep expertise:

1. **Build a dbt (Data Build Tool) layer:** Don't just write queries; build a documented pipeline that tests for data quality (e.g., "Ensure no lap time is negative" or "Ensure energy used doesn't exceed battery capacity").
2. **Address the "Cold Start" Problem:** How would you model a brand-new track (like a new street circuit in LA) using only historical data from similar street circuits? This requires **feature engineering** (grading tracks by "Aggression Level" or "Traction Demand").
3. **Stakeholder Presentation:** Create three different views for your final project:
 - **The Technical View:** (For TRI Engineers) The Python code and simulation accuracy.
 - **The Business View:** (For Toyota HQ) How Formula E success correlates with brand "coolness" and sales in EV-heavy markets.
 - **The Product View:** (For UX Designers) What data from the race car should be shown on a consumer car's dashboard to help drivers save energy

What's excellent about this spec:

The star schema + dbt layer combination is exactly the kind of infrastructure thinking that impresses data engineering-adjacent analyst roles. The "Toyota Efficiency Score" as a proprietary metric shows business intuition, not just SQL skills. And the California simulation bridges racing data to a real market problem, which is the kind of cross-domain thinking TRI would respond to.

Where we need to be careful:

A few pieces of the spec are aspirationally scoped in ways that could create problems:

- **The Formula E energy consumption per lap;** this data isn't publicly available at the granularity described. Formula E doesn't publish lap-level energy telemetry. We can work around this with what *is* available (lap times, gap data, race results from the FIA/Formula E API or scraping), but we'll need to be honest in the write-up about where we're modeling vs. measuring.
- **"Regen efficiency" data,** same issue. We'll need to engineer a proxy metric from available data rather than use direct telemetry.

- **dbt**, totally buildable, but it adds real setup time. Worth including if you want this to signal data engineering chops specifically.